Countable Additivity is Not First-Order Axiomatizable

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Abstract

In the first-order language of Boolean algebras equipped with a normalized finitely additive charge, no first-order theory characterizes those models whose charge extends to a σ-additive measure on the generated σ-algebra. Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras. The proof uses a single ultraproduct construction: Dirac charges on the finite-cofinite algebra are individually σ-additive, but their ultraproduct is the finite-cofinite charge, which fails σ-additivity — contradicting Łoś's theorem if any first-order axiomatization existed. As an application, the same obstruction shows that collective exhaustion — the condition requiring that whenever a decreasing sequence of observable events has empty intersection, the assigned charges converge to zero — cannot be derived from any first-order structural condition, even though it is the necessary and sufficient condition for σ-additive extension in a directed system of Boolean algebras.

1 Introduction

A classical question in model theory is whether a mathematically natural property of structures can be captured by a first-order theory in an appropriate language. When the property is preserved under ultraproducts, Łoś's theorem ensures it is at least consistent with first-order axiomatization; when it fails in an ultraproduct of structures that individually satisfy it, first-order axiomatization is ruled out.

Countable additivity is a fundamental property of probability measures, distinguishing σ-additive measures from merely finitely additive charges. It is well known that countable additivity cannot be expressed by a single first-order sentence, since it involves quantification over countable sequences. The stronger question — whether countable additivity is *axiomatizable* by any first-order theory, i.e., whether there exists a set of first-order sentences whose models are exactly the σ-additive ones among finitely additive probability Boolean algebras — appears not to have been explicitly settled in the literature in the natural language we consider here.

We settle this question negatively. In the first-order language $\mathcal{L}_{\mathrm{BA},\mu}$ of Boolean algebras with a normalized finitely additive charge (Section 2), no first-order theory characterizes those models whose charge extends to a σ-additive measure. The proof is a direct ultraproduct argument using the finite-cofinite algebra on $\mathbb{Q}$ as the canonical witness (Section 3).

Section 4 draws out an application to collective exhaustion (CE), the necessary and sufficient admissibility condition for σ-additive extension in directed systems of Boolean algebras. Section 5 raises the representation question of which purely finitely additive charges arise as ultraproducts of σ-additive ones, which is strictly harder than non-axiomatizability and is left open.

2 The language

Let $\mathcal{L}_{\mathrm{BA},\mu}$ be the one-sorted first-order language with:

- Boolean operations $\wedge, \vee, \neg, 0, 1$;

- for each rational $q \in [0,1] \cap \mathbb{Q}$, unary relation symbols $M_{\leq q}(x)$ and $M_{\geq q}(x)$, intended to mean $\mu(x) \leq q$ and $\mu(x) \geq q$ respectively.

A structure for $\mathcal{L}_{\mathrm{BA},\mu}$ is a Boolean algebra B together with a $[0,1]$-valued finitely additive probability μ, encoded through the truth of the predicates $M_{\leq q}$ and $M_{\geq q}$. Using rational comparison predicates rather than a function symbol into $[0,1]$ avoids introducing a second numeric sort.

Definition 2.1 (First-order theory of finitely additive probabilities)**.** Let T_{fa} be the $\mathcal{L}_{\mathrm{BA},\mu}$-theory axiomatizing:

(i) B is a Boolean algebra;

(ii) normalization: $\mu(0) = 0$ and $\mu(1) = 1$;

(iii) monotonicity: $x \leq y \Rightarrow \mu(x) \leq \mu(y)$;

(iv) finite additivity: for all rationals r, s with $r + s \leq 1$,

$$x \wedge y = 0 \wedge M_{\leq r}(x) \wedge M_{\leq s}(y) \;\Rightarrow\; M_{\leq r+s}(x \vee y),$$

and similarly for lower bounds.

The class of Boolean algebras with normalized finitely additive probability is first-order axiomatizable in $\mathcal{L}_{\mathrm{BA},\mu}$. Countable additivity, by contrast, requires quantification over an arbitrary countable sequence $(a_n)_{n\in\mathbb{N}}$ — not available in first-order logic — and so is not a sentence of $\mathcal{L}_{\mathrm{BA},\mu}$.

3 The main theorem

Proposition 3.1 (σ-additive extension is not first-order axiomatizable)**.** *There is no first-order $\mathcal{L}_{\mathrm{BA},\mu}$-theory $T \supseteq T_{\mathrm{fa}}$ such that, for every $(B,\mu) \models T_{\mathrm{fa}}$,*

$$(B,\mu) \models T \quad\Longleftrightarrow\quad \mu \textit{ extends to a } \sigma\textit{-additive measure on } \sigma(B).$$

Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras.

Proof. Let E be the Boolean algebra of finite-cofinite subsets of $\mathbb{Q}$, fix an enumeration $q : \mathbb{N} \to \mathbb{Q}$, and let δ_n be the Dirac charge at $q(n)$. Each (E, δ_n) is a model of T_{fa} in which δ_n is σ-additive. Suppose for contradiction that a theory T as described exists; then each $(E, \delta_n) \models T$.

Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$ extending the cofinite filter. By Łoś's theorem [1], the ultraproduct $\prod_{\mathcal{U}}(E, \delta_n)$ also models T. The charge induced on the diagonal copy of E is

$$\ell(A) = \lim_{\mathcal{U}} \delta_n(A) = \lim_{\mathcal{U}} \mathbf{1}_{q(n)\in A}.$$

If A is finite, $q(n) \in A$ for only finitely many n, so $\ell(A) = 0$. If A is cofinite, $q(n) \in A$ for cofinitely many n, so $\ell(A) = 1$ since $\mathcal{U}$ extends the cofinite filter. Thus ℓ is the finite-cofinite charge.

The sequence $A_k = \mathbb{Q} \setminus \{q(0), \ldots, q(k)\}$ satisfies $A_0 \supseteq A_1 \supseteq \cdots$ and $\bigcap_k A_k = \varnothing$, yet $\ell(A_k) = 1$ for all k. Hence the charge induced on the diagonal copy of E fails σ-additivity and does not extend to a σ-additive measure on $\sigma(E)$. Consequently the ultraproduct cannot satisfy any theory T characterizing σ-additive extension, contradicting Łoś's theorem applied to the assumption that each $(E, \delta_n) \models T$. □

Remark 3.2. For precision: the ultraproduct $\prod_{\mathcal{U}}(E, \delta_n)$ is an $\mathcal{L}_{\mathrm{BA},\mu}$-structure in its own right; ℓ is the charge induced on the diagonal copy of E inside it. Identifying the ultraproduct with (E, ℓ) is harmless for the contradiction, since the relevant failure of σ-additivity is witnessed on the diagonal copy.

4 Application to collective exhaustion

Let $(\mathcal{E}_i, \ell_i)_{i\in\iota}$ be a directed system of Boolean charge spaces: each $\mathcal{E}_i$ is a Boolean algebra and $\ell_i : \mathcal{E}_i \to [0,1]$ is a normalized finitely additive charge, with compatible connecting maps. *Collective exhaustion* (CE) is the following condition on the induced charge ℓ on the direct limit algebra $\mathcal{E} = \varinjlim \mathcal{E}_i$: whenever $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}$ with $\bigcap_n E_n = \varnothing$, we have $\ell(E_n) \to 0$.

CE is a valuation-layer condition: it constrains how charges behave across the directed system, not the Boolean-algebraic structure of the index set. It is the necessary and sufficient condition under which a compatible finitely additive family extends to a σ-additive measure on the observable σ-algebra generated by the system.

Proposition 3.1 implies immediately:

Corollary 4.1 (CE is not first-order)**.** *In the first-order language $\mathcal{L}_{\mathrm{BA},\mu}$ of Boolean algebras with normalized finitely additive charge, no first-order $\mathcal{L}_{\mathrm{BA},\mu}$-theory characterizes those models satisfying CE. Equivalently, no accumulation of first-order conditions on the Boolean-algebraic structure can imply collective exhaustion.*

The finite-cofinite charge of Proposition 3.1 is the canonical witness: it satisfies every first-order structural condition yet assigns persistent unit mass to the decreasing sequence $A_k = \mathbb{Q} \setminus \{q(0), \ldots, q(k)\}$, whose intersection is empty. CE fails precisely because the mass is not exhausted.

Foundational disputes about probability are not terminological. They concern which extra structure licenses the passage from finite coherence to probability, and Corollary 4.1 shows that some such structure is logically unavoidable: no first-order condition can close the gap. For a full development of CE as the admissibility condition for σ-additive extension in a directed observational system, see [2].

5 The representation question

Proposition 3.1 shows that *one* purely finitely additive charge arises as an ultraproduct of σ-additive ones — that single witness suffices for the non-axiomatizability conclusion. A stronger question is:

> *Which purely finitely additive charges arise as ultraproducts or ultralimits of σ-additive probabilities?*

This is a representation question, and it is strictly harder than non-axiomatizability. It may hold only after enlarging the notion of approximation, for example by allowing nets or ultralimits more general than literal ultraproducts. The Yosida–Hewitt decomposition [3] identifies the purely finitely additive part ℓ_p of any finitely additive charge; whether ℓ_p can always be represented as an ultralimit is not settled here and is left as an open question.

References

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[3] Kôsaku Yosida and Edwin Hewitt. Finitely additive measures. *Transactions of the American Mathematical Society*, 72(1):46–66, 1952.